

1.1.7 Universal & Advanced Logic Gates

NAND کا مطلب ہے NOT AND اور NOR کا مطلب ہے NOT OR

CS XI

Chapter 1: Computer Systems

Lecture 05 of 12

1.1.7 Universal & Advanced Logic Gates

LEARN IT

NAND means "NOT AND": it is an AND gate followed by a NOT. **NOR** means "NOT OR": an OR gate followed by a NOT. On the symbol, the NOT is shown as a small bubble at the output.

NAND and NOR are called **universal gates** because **any logic circuit in the world can be built using only NAND gates (or only NOR gates)**. Factories like this: they can mass-produce one type of chip and build everything from it.



$$\text{NAND: } Y = (A \cdot B)'$$



$$\text{NOR: } Y = (A + B)'$$

NAND = AND symbol + bubble. NOR = OR symbol + bubble.

NAND (NOT AND) — $Y = (A \cdot B)'$

A	B	A·B	Y
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

NOR (NOT OR) — $Y = (A + B)'$

A	B	A+B	Y
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

WORKED EXAMPLE

Show that a NAND gate can work as a NOT gate.

Step 1: Connect the same input A to both inputs of the NAND gate. Then $Y = (A \cdot A)'$.

Step 2: Since $A \cdot A = A$, we get $Y = A'$.

Step 3: Check: $A = 0 \rightarrow (0 \cdot 0)' = 0' = 1$. $A = 1 \rightarrow (1 \cdot 1)' = 1' = 0$. This is exactly the NOT truth table.

In the same way, a NOR gate with both inputs tied together gives $(A + A)' = A'$. And a NAND followed by a NAND-as-NOT gives $((A \cdot B)')' = A \cdot B$, an AND gate. This is why they are "universal".

REMEMBER IT

NAND is the exact **opposite** of AND (0 only at 11). NOR is the exact opposite of OR (1 only at 00). Just flip the AND/OR output column.

COMMON MISTAKE

Do not forget the bubble. Without the small circle, a NAND symbol becomes an AND symbol and your answer is wrong. Also, $(A \cdot B)'$ is **not** the same as $A' \cdot B'$: for $A = 1, B = 0$, $(1 \cdot 0)' = 1$ but $0 \cdot 1 = 0$.

اردو میں سمجھیں

NAND کا مطلب ہے NOT AND اور NOR کا مطلب ہے NOT OR. انہیں یونیورسل (عالمگیر) گیٹ کہتے ہیں کیونکہ صرف NAND یا صرف NOR گیٹس سے دنیا کا کوئی بھی لاجک سرکٹ بنایا جا سکتا ہے۔
NAND صرف تب 0 دیتا ہے جب دونوں ان پٹ 1 ہوں، اور NOR صرف تب 1 دیتا ہے جب دونوں ان پٹ 0 ہوں۔

CHECK YOURSELF

1. Why are NAND and NOR called universal gates?
2. What is the NOR output for $A = 0, B = 1$?
3. How do you make a NOT gate from a NAND gate?

Answers: 1. Any logic circuit can be built using only NAND gates or only NOR gates. 2. 0. 3. Join both inputs together and feed A into them: $(A \cdot A)' = A'$.

LEARN IT

XOR (Exclusive OR) gives output 1 when the inputs are **different**. Expression: $Y = A \oplus B$. In terms of basic operations, $A \oplus B = A'B + AB'$.

XNOR (Exclusive NOR) is XOR followed by NOT. It gives output 1 when the inputs are the **same**. Expression: $Y = (A \oplus B)' = A'B' + AB$.

The XOR symbol is an OR symbol with an extra curved line at the input side. XNOR adds a bubble at the output.



$$\text{XOR: } Y = A \oplus B$$



$$\text{XNOR: } Y = (A \oplus B)'$$

XOR = OR symbol with an extra curve at the back. XNOR = XOR symbol + bubble.

XOR — $Y = A \oplus B$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

XNOR — $Y = (A \oplus B)'$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

All seven gates at a glance

A	B	AND	OR	NAND	NOR	XOR	XNOR
0	0	0	0	1	1	0	1
0	1	0	1	1	0	1	0
1	0	0	1	1	0	1	0
1	1	1	1	0	0	0	1

(NOT has one input: $A = 0 \rightarrow 1$, $A = 1 \rightarrow 0$.)

WORKED EXAMPLE

A staircase light. A light has one switch at the bottom (A) and one at the top (B) of a staircase. Flipping either switch should change the light.

Step 1: Start with both switches down: $A = 0, B = 0$, light off ($Y = 0$).

Step 2: Flip the bottom switch: $A = 1, B = 0 \rightarrow$ light on ($Y = 1$).

Step 3: Walk up and flip the top switch: $A = 1, B = 1 \rightarrow$ light off ($Y = 0$).

Step 4: The pattern 0, 1, 1, 0 is the XOR truth table. So the light is $Y = A \oplus B$.

REMEMBER IT

Memory trick: XOR outputs 1 when inputs are "eXclusively" different. XNOR outputs 1 when inputs are the **same**. XNOR is sometimes called the "equality gate".

COMMON MISTAKE

XOR is not the same as OR. They differ only in the last row: for $A = 1, B = 1$, OR gives 1 but XOR gives 0.

اردو میں سمجھیں

XOR گیٹ تب 1 دیتا ہے جب دونوں ان پٹ مختلف ہوں، اور XNOR گیٹ تب 1 دیتا ہے جب دونوں ان پٹ ایک جیسے ہوں۔ XNOR دراصل XOR کے بعد NOT ہے۔ OR اور XOR میں فرق صرف آخری قطار میں ہے جہاں دونوں ان پٹ 1 ہوں۔

CHECK YOURSELF

1. What is $1 \oplus 1$?
2. Which gate outputs 1 when both inputs are the same?
3. Write XOR using AND, OR and NOT.

Answers: 1. 0. 2. XNOR. 3. $A \oplus B = A'B + AB'$.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. NAND and NOR are universal because:

- a. Any circuit can be built from only that gate
- b. They are analog
- c. They store files
- d. They sort arrays

MCQ 2. NAND is:

- a. NOT of AND
- b. AND of NOT
- c. OR of XOR
- d. Only OR

MCQ 3. NAND of 1,1 is:

- a. 0
- b. 1
- c. Z
- d. 2

MCQ 4. NOR of 0,0 is:

- a. 1
- b. 0
- c. A
- d. B

MCQ 5. XOR is 1 when inputs are:

- a. Different
- b. Both 1 only
- c. Both 0 only
- d. Always 1

MCQ 6. XNOR is 1 when inputs are:

- a. The same
- b. Different
- c. Only 1,0
- d. Floating

Extra short questions

1. Why are NAND and NOR called universal gates?
2. Write NAND and NOR expressions.
3. Memory trick for XOR vs XNOR.
4. Truth table of XOR.

Extra long question

1. Show how a NOT, AND and OR can each be built from NAND only. Include truth-table checks.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Any Boolean function (AND, OR, NOT, XOR...) can be built using only NAND, or only NOR.
2. $Y = (A \cdot B)'$ and $Y = (A + B)'$.
3. XOR: eXclusively different $\rightarrow 1$. XNOR: same $\rightarrow 1$ (equivalence).
4. $00 \rightarrow 0$, $01 \rightarrow 1$, $10 \rightarrow 1$, $11 \rightarrow 0$.

Long

1. NOT: NAND with inputs tied. AND: NAND then NAND-NOT. OR: invert inputs with NAND-NOTs then NAND (De Morgan). Tables must match the target gate.

Next lecture: 1.1.8 – 1.1.11 Logic Expressions, Minterms & Diagrams